

FORMALISING THE HEISENBERG UNCERTAINTY PRINCIPLE

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INTRODUCTION

In 1925, Werner Heisenberg’s *Umdeutung* paper, fundamentally reinterpreting the heated debate on the developing theory of Quantum Mechanics and rendering obsolete the classical approach of Niels Bohr to atomic theory, was published. In the words of Paul Dirac [Dir77, Lecture 1. The Development of Quantum Mechanics], Heisenberg “had the idea that physical theory should concentrate on quantities which are closely related to observed quantities. Now, the things you observe are only very remotely connected with the Bohr orbits. So Heisenberg said that the Bohr orbits are not very important. The things that are observed, or which are connected closely with the observed quantities, are all associated with two Bohr orbits and not with just one Bohr orbit: two instead of *one*”. The effect is that one needs a matrix to store all the quantities associated to a couple of orbits, and this is the foundational idea behind the so-called *matrix mechanics* formalism that was developed. In reality, Heisenberg simply postulated a product rule for transition amplitudes $\mathcal{A}(o_1, o_2)$ between orbit o_1 and o_2 . It was Max Born who first recognised the well-known form of a matrix product and understood the significance of this, mainly because product of physically observable quantities $\mathcal{A}, \mathcal{B}$ (such as an amplitude) were thought to be naturally commuting, for the displeasure of many physicist like Wolfgang Pauli, who accused Born of being “fond of tedious and complicated formalisms. You are only going to spoil Heisenberg’s physical ideas by your futile mathematics” [Bor14].

In matrix mathematics, there are thus pairs of variables that don’t commute, such as position and momentum or energy and time, and for such observables an “uncertainty principle” holds in the following sense: for instance, the shorter the time period t during which a magnetic field acts on an atom, the less accurate the energy measurement E . [Meh01, Chapter II]

He then made this principle quantitative: given observables $\mathcal{A}, \mathcal{B}$ satisfying the commutation relation

$$0 \neq \mathcal{A}\mathcal{B} - \mathcal{B}\mathcal{A} \propto i\hbar$$

then the *uncertainty* $\sigma_{\mathcal{A}}, \sigma_{\mathcal{B}}$ in the measurement of these observables satisfies:

$$\sigma_{\mathcal{A}}\sigma_{\mathcal{B}} \geq \frac{1}{2}. \tag{H}$$

We provide here a full formalisation in Lean of the functional-analytic proof of this fact, that is the argument presented by in the monograph of Peter D. Lax on Functional Analysis [Lax14] and falling into the theoretical framework introduced by von Neumann for the formulation of Quantum Mechanics using his theory of Hilbert spaces.

1. QUANTUM STATES, OBSERVABLES AND THEIR MEASUREMENTS

A quantum state $|\psi\rangle$ is taken to be a unit vector in some complex Hilbert space $\mathbf{H}$, with norm

$$1 = \|\psi\|^2 = \langle\psi|\psi\rangle.$$

Here, $\langle\psi| \in \mathbf{H}^*$ denotes the duality pairing and the notation for the inner product $\langle\psi|\psi\rangle$, introduced by Dirac, makes implicit use of the Fréchet-Riesz representation theorem, prescribing an inner product representation for the action of such pairing. In finite-dimensional Euclidean spaces, quantum states and their pairing respectively correspond to column and row vectors. In Lean, we fix a quantum state that we shall use throughout:

```
variable {ψ : H}
variable (h_QuantumState : ‖ψ‖ = 1)
```

as all our statements will hold pointwise. Quantum Mechanics describes the measurements of observables associated to such states, here represented by self-adjoint operators $\mathcal{A}$ over $\mathbf{H}$:

```
variable (A : H →L[C] H)
variable (hA : A.IsSymmetric)
```

though in Lean the assumption of the operator being symmetric:

$$\langle\psi_1|\mathcal{A}\psi_2\rangle = \langle\mathcal{A}\psi_1|\psi_2\rangle \quad (\dagger)$$

is enough, since our quantum state is fixed and we implicitly assume it is contained in the domain of the operator considered*. Most importantly, it plays no role in the proof.

This formalism is useful to study, for instance, measurement in the energy of a quantum systems: $\mathcal{H}|\psi\rangle = E|\psi\rangle$ so that the discrete energy spectrum prescribed by the theory are the eigenvalues of the Hamiltonian operator $\mathcal{H}$ and are all multiples of Planck's constant $\hbar$. The interpretation of `h_QuantumState` is the Born rule: think of $\mathbf{H}$ as the space of square-integrable functions, then the square magnitude of the solutions to the Schrödinger equation $\mathcal{H}|\psi\rangle = E|\psi\rangle$, postulated in a way that the elliptic differential operator $\mathcal{H}$ reduces to the Hamilton-Jacobi operator from Classical Mechanics as $\hbar \sim 0$, give the probability density of the observable measured, thus it must integrate to 1.

This also justifies the definition of the *measurement*, or *expected value*, $a = \langle\psi|\mathcal{A}|\psi\rangle$ associated to the given observable $\mathcal{A}$, i.e. in Lean:

```
noncomputable def Measurement (A : H →L[C] H) (ψ : H) : ℂ := ⟨ψ | A ψ⟩
variable (a : ℂ) (a_def : a = Measurement A ψ)
```

*The concept came in fact much later and was of almost no interest to physicist, so that when someone “pointed out that it was a mathematician, von Neumann, who clarified the difference between a self-adjoint operator and one that is merely symmetric, «What’s the difference», replied Heisenberg.” [Lax14, Chapter 33]

We prove that this is a real number, in fact

$$\bar{a} = \overline{\langle \psi | \mathcal{A} \psi \rangle} = \langle \mathcal{A} \psi | \psi \rangle \stackrel{(\dagger)}{=} \langle \psi | \mathcal{A} \psi \rangle = a,$$

using the `inner_conj_symm` Mathlib4 lemma:

```
inner_conj_symm.{u_1, u_2} {F : Type u_1} {H : Type u_2}
[RCLike F] [SeminormedAddCommGroup H] [InnerProductSpace F H]
(x y : H) : (starRingEnd F) ⟨y, x⟩_F = ⟨x, y⟩_F
```

to rewrite the conjugate of the inner product and `LinearMap.IsSymmetric.apply_clm` to formalise $(\dagger)$. It follows that $\|a\|^2 = a \cdot \bar{a} = a^2$ and that the imaginary part of a is null. Also, I was having serious problems when matching the types of $a \in \mathbb{C}$ and real-valued objects such as norms. It was suggested to me in a Lab, by Professor Mehta, to prove the following result

```
lemma meas_repr : ∃ A_meas : ℝ, ⟨ψ | A ψ⟩ = A_meas
```

so that I could, when convenient, obtain the real-valued representation of the measurement.

2. UNCERTAINTY

The *uncertainty* $\sigma_{\mathcal{A}}$ in the measurement of an observable $\mathcal{A}$ in state $|\psi\rangle$ is the square root of the expected value in the state $|\psi\rangle$ of $(\mathcal{A} - a\mathcal{I})^2$, where $\mathcal{I}$ is the identity operator, i.e.

```
noncomputable def Uncertainty (A : H → L[ℂ] H) (ψ : H) : ℂ :=
  let obs : H → L[ℂ] H := A - a • 1
  let var : H → L[ℂ] H := obs * obs
  Measurement var ψ ^ (2⁻¹ : ℂ)
```

Notice that the operator $\mathcal{A} - a\mathcal{I}$ is still symmetric, in fact:

```
(A - a • 1).IsSymmetric := by
calc ⟨A u - a • u | v⟩ = ⟨A u | v⟩ - ⟨a • u | v⟩ := by rw [inner_sub_left]
_ = ⟨u | A v⟩ - ⟨a • u | v⟩ := by simp only [LinearMap.IsSymmetric.apply_clm (hA)]
_ = ⟨u | A v⟩ - a * ⟨u | v⟩ := by rw [inner_smul_left, meas_real a A hA a_def]
_ = ⟨u | A v⟩ - ⟨u | a • v⟩ := by rw [← inner_smul_right]
_ = ⟨u | A v - a • v⟩ := by rw [← inner_sub_right]
```

where `meas_real` is the lemma we have proved that $\bar{a} = a$, the `inner_sub_left` and `inner_sub_right` results formalise biadditivity of the inner product, while `inner_smul_left` and `inner_smul_right` respectively formalise the sesquilinearity identities:

$$\langle \psi_1 | a \psi_2 \rangle = a \langle \psi_1 | \psi_2 \rangle, \quad \langle a \psi_1 | \psi_2 \rangle = \bar{a} \langle \psi_1 | \psi_2 \rangle, \quad a \in \mathbb{C}.$$

We thus prove the `uncertainty_squared` formula

$$\text{Uncertainty } a \, A \, \psi \, ^2 = \|A \, \psi\|^2 - a^2 \, * \, \|\psi\|^2$$

by symmetry as follows:

$$\begin{aligned} \sigma_{\mathcal{A}}^2 &\equiv \langle \psi | (\mathcal{A} - a\mathcal{I})^2 | \psi \rangle \stackrel{(\dagger)}{=} \langle \mathcal{A}\psi - a\psi | \mathcal{A}\psi - a\psi \rangle \\ &= \langle \mathcal{A}\psi | \mathcal{A}\psi \rangle - \langle a\psi | \mathcal{A}\psi \rangle - \langle \mathcal{A}\psi | a\psi \rangle + \langle a\psi | a\psi \rangle \\ &= \|\mathcal{A}\psi\|^2 - \bar{a}\langle \psi | \mathcal{A}\psi \rangle - a\langle \mathcal{A}\psi | \psi \rangle + a^2\|\psi\|^2 \\ &\stackrel{(\dagger), \text{meas_real}}{=} \|\mathcal{A}\psi\|^2 - 2a^2 + a^2\|\psi\|^2 \end{aligned}$$

and we conclude by `h_QuantumState`. In particular,

$$\sigma_{\mathcal{A}} = 0 \quad \Longleftrightarrow \quad \mathcal{A}\psi = a\psi$$

and thus we have the following:

```
theorem uncertainty_iff_eigenstate
  (h_QuantumState : ‖ψ‖ = 1) (hA : A.IsSymmetric) (a_def : a = Measurement A ψ) :
  Uncertainty a A ψ = 0 ↔ Module.End.HasEigenvector (A : H →ℂ ℂ) H) a ψ
```

Both directions are almost trivial and follow from `Module.End.mem_eigenspace_iff`. These last two results I formalised for completeness, but they are of no use in the proof of our objective, though they provide intuition on why (H) makes sense.

According to `uncertainty_iff_eigenstate`, given a pair of observables $\mathcal{A}, \mathcal{B}$, both can be measured with absolute certainty in the same state $|\psi\rangle$ if and only if $|\psi\rangle$ is an eigenvector of both $\mathcal{A}$ and $\mathcal{B}$. Pairs of commuting operators have common eigenvectors, but in general, this is not to be expected. [Lax14]

3. THE UNCERTAINTY PRINCIPLE

We now consider a pair of, not necessarily commuting, observables $\mathcal{A}, \mathcal{B}$ and look at their commutator:

```
def Commutator (A B : H →ℂ ℂ) H) : H →ℂ ℂ) H := A * B - B * A
```

which we shall denote by $[\mathcal{A}, \mathcal{B}]$. This can be seen as a measure of how much the two observables do *not* commute: for instance, an observable $\mathcal{A}$ always commutes with itself and thus:

```
example : Commutator A A = 0 := by rw [Commutator]; simp
```

We impose the commutation relation

$$[\mathcal{A}, \mathcal{B}] = \mathcal{A}\mathcal{B} - \mathcal{B}\mathcal{A} = i\mathcal{I}$$

so that the two observables have no common eigenvector, for such a vector would be mapped into zero by $\mathcal{I}$, showing that such observables cannot both be measured with absolute certainty. This uncertainty is, as we said, quantified by (H), which we now prove.

We start proving a weak reformulation of the commutation relation, namely

$$\langle \mathcal{A}\psi | \mathcal{B}\psi \rangle - \langle \mathcal{B}\psi | \mathcal{A}\psi \rangle = i\|\psi\|^2.$$

Proof. $\langle \mathcal{A}\psi | \mathcal{B}\psi \rangle - \langle \mathcal{B}\psi | \mathcal{A}\psi \rangle = \langle \psi | \mathcal{A}\mathcal{B}\psi \rangle - \langle \psi | \mathcal{B}\mathcal{A}\psi \rangle = \langle \psi | \mathcal{A}\mathcal{B} - \mathcal{B}\mathcal{A} | \psi \rangle = i\langle \psi | \psi \rangle = i\|\psi\|^2.$ $\square$

This, we formalise in Lean using a `calc` block with the same Mathlib results used before. A new lemma we instead use in the proof of another preliminary result, the inequality:

$$|\langle \psi | \mathcal{A}\psi + it\mathcal{B}\psi \rangle|^2 \leq \|\mathcal{A}\psi + it\mathcal{B}\psi\|^2, \quad t \in \mathbb{R} \quad (\text{cauchy_schwarz})$$

relying on the Cauchy-Schwarz inequality:

$$|\langle \psi | \mathcal{A}\psi + it\mathcal{B}\psi \rangle|^2 \leq \|\psi\|^2 \cdot \|\mathcal{A}\psi + it\mathcal{B}\psi\|^2$$

and `h_QuantumState`.

Letting $b = \langle \psi | \mathcal{B}\psi \rangle$, then it holds:

$$a^2 + t^2 b^2 = |\langle \psi | \mathcal{A}\psi + it\mathcal{B}\psi \rangle|^2$$

and consequently the most important inequality

$$a^2 + t^2 b^2 \leq \|\mathcal{A}\psi\|^2 - t + t^2 \|\mathcal{B}\psi\|^2 \quad (\text{meas_ineq})$$

in fact:

$$\begin{aligned} a^2 + t^2 b^2 &= |\langle \psi | \mathcal{A}\psi + it\mathcal{B}\psi \rangle|^2 \stackrel{(\text{cauchy_schwarz})}{\leq} \|\mathcal{A}\psi + it\mathcal{B}\psi\|^2 = \langle \mathcal{A}\psi + it\mathcal{B}\psi | \mathcal{A}\psi + it\mathcal{B}\psi \rangle \\ &= \langle \mathcal{A}\psi | \mathcal{A}\psi \rangle + \langle \mathcal{A}\psi | it\mathcal{B}\psi \rangle + \langle it\mathcal{B}\psi | \mathcal{A}\psi \rangle + \langle it\mathcal{B}\psi | it\mathcal{B}\psi \rangle \\ &= \|\mathcal{A}\psi\|^2 + it[\langle \mathcal{A}\psi | \mathcal{B}\psi \rangle - \langle \mathcal{B}\psi | \mathcal{A}\psi \rangle] + t^2 \|\mathcal{B}\psi\|^2 \end{aligned}$$

and we conclude using the weak formulation of the Heisenberg commutation relation. This was the most involved proof to formalise, because it is here that I realised the problems I was having due to the mismatch in the types of $a, b \in \mathbb{C}$ and the real-valued right-hand side, even though at the end it turned out to be a rather simple `calc` block with basically the same results used above. Now the proof is basically concluded as, subtracting the left-hand side to `(meas_ineq)`,

$$0 \leq (\|\mathcal{A}\psi\|^2 - a^2) - t + t^2 (\|\mathcal{B}\psi\|^2 - b^2) = \sigma_{\mathcal{A}}^2 - t + t^2 \sigma_{\mathcal{B}}^2 \quad (\text{ineq})$$

i.e. the quadratic polynomial on the right is nonnegative for all $t \in \mathbb{R}$ and thus its discriminant is nonpositive:

$$1 - 4\sigma_{\mathcal{A}}^2 \sigma_{\mathcal{B}}^2 \leq 0,$$

this being the Mathlib result `discrim_le_zero`. Since our t was fixed, we need just to generalise the nonnegativity statement first:

```

∀ t : ℝ, 0 ≤ (σB ^ 2).re * t^2 - t + (σB ^ 2).re := by
  intro t
  have h_ineq := ineq A a B b hA hB t h_QuantumState a_def b_def heisenberg
  have h_sq : t ^ 2 = t * t := by ring
  rw [h_sq] at h_ineq
  linarith

```

and we are done.

DIFFICULTIES AND CONCLUSIONS

The proof to be formalised literally fits in one page, however many steps that a human mathematician's eye accepts without blinking require a lot of more care in Lean, the byproduct of this being all the preliminary results I had to prove and that Lax takes for granted. After all, the number of results from Mathlib used is little and repeated: the biggest difficulty was in fact related to setting up the right “environment” of definitions and mathematical structures so that performing operations with inner products and operators would be smooth. I struggled a bit with this and I ended giving up the first attempt I made, defining actual structures like as follows:

```

structure QuantumState where
  ψ : H
  is_unit : ‖ψ‖ = 1

local notation "<" state1 "|" state2 ">" => inner ℂ state1 state2

structure Observable where
  domain : Submodule ℂ H
  A : domain →[ℂ] H
  self_adjoint : ∀ (ψ φ : domain), ⟨A ψ | ↑φ⟩ = ⟨↑ψ | A φ⟩

```

While more general, allowing to prove the uncertainty principle for potentially unbounded operators, this approach felt very cumbersome as, if I for instance wanted to compute something related to the quantum state $|\psi\rangle$, I had to access it via `QuantumState.ψ` and similarly the actual operators would have been `Observable.A`, which was not convenient. I tried to solve this, also asking Google Gemini for help in this conversation, being suggested to coerce structures like as follows:

```

instance : Coe (QuantumState H) (H) where coe state := state.ψ

```

The problem with this was that I had to coerce the action of observables on states as follows:

```
noncomputable instance : CoeFun (Observable H) (fun C => H → H) where
  coe obs ψ := if h : ψ ∈ obs.domain then obs.A ⟨ψ, h⟩ else 0
```

i.e. setting $\mathcal{A}|\psi\rangle$ to zero if $|\psi\rangle$ was not in the domain of the operator. However, the very first step of the very first proof I attempted to formalise using this approach was intractable:

```
local notation "σ" A => Uncertainty H a A ψ
theorem uncertainty_squared (h : (ψ : H) ∈ A.domain) :
  (σ A)^2 = ‖A ψ‖^2 - a^2 := by
  rw [Uncertainty]
  sorry
```

ending up in the following goal:

```
⊢ (⟨ψ.ψ | (fun ψ ↦
  (fun ψ ↦ (fun ψ ↦ if h : ψ ∈ A.domain then A.A ⟨ψ, h⟩ else 0) ψ - a • ψ)
  ((fun ψ ↦ (fun ψ ↦ if h : ψ ∈ A.domain then A.A ⟨ψ, h⟩ else 0) ψ - a • ψ) ψ))
  ψ.ψ⟩ ^ 1 / 2) ^ 2 =
  ↑ ‖(fun ψ ↦ if h : ψ ∈ A.domain then A.A ⟨ψ, h⟩ else 0) ψ.ψ‖ ^ 2 - a ^ 2
```

As explained above, subtleties about the domain of the operators, and thus the distinction between self-adjoint and symmetric ones, do not play a role in the proof, so I decided to fix $|\psi\rangle$ and treat it as always contained in the domain of the operators considered.

I was suggested in a Lab, by Professor Buzzard, that it would have been a good idea to let the type of `Measurement` and `Uncertainty` to be a real number from the beginning, in a fashion similar to the following:

```
noncomputable def Measurement (A : H → L[ℂ] H) (ψ : H) : ℝ := ⟨ψ | A ψ⟩.re
```

I was kind of convinced and I started trying it out in a new file: however, this did not fix much as I still encountered object of type `ℂ`, for instance in the proofs of `obs_symmetric`, `uncertainty_squared` and `weak_heisenberg`, involving expressions like $\langle \mathcal{A}\psi | \mathcal{B}\psi \rangle$ which are not necessarily real. Namely, I was trading off a mismatch between real and complex numbers with a mismatch between complex and real numbers!

Thus, I decided to stick with the complex-typed approach.

The only other major obstacle was more related to my understanding of how the informal proof is written by Lax. In particular, his weakened Heisenberg commutation relation looks like:

$$\langle \mathcal{B}\psi | \mathcal{A}\psi \rangle - \langle \mathcal{A}\psi | \mathcal{B}\psi \rangle = i\|\psi\|^2.$$

The mismatch is due to the convention adopted for the sesquilinearity identities, as I came to realise also with the aid, again, of Google Gemini in this chat.

In conclusion, I have greatly enjoyed the fact that one can make formal something which is not really a theorem, since the Uncertainty Principle is after all regarded as a fundamental law of modern physics. It would be wrong to reduce it to mere properties of operators acting on Hilbert spaces, and that is why I found this formalisation very interesting to carry out.

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